
Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix, spanning hydropower, wind, and solar technologies.

Source: Our World in Data — Renewable Energy
Data: Ember (2026); Energy Institute (2025)



~30%

of global electricity
now from renewables

75% of Global GHG Emissions Still Come from Fossil Fuels

Since the Industrial Revolution, fossil fuels have dominated the global energy system. The consequences are clear — and urgent.

75%

of global greenhouse gas
emissions from fossil fuels

5M+

premature deaths per year
from air pollution

170+

years of fossil fuel
dominance since Industrial
Revolution

To reduce CO₂ emissions and air pollution, the world must rapidly shift toward low-carbon sources — nuclear and renewable technologies.

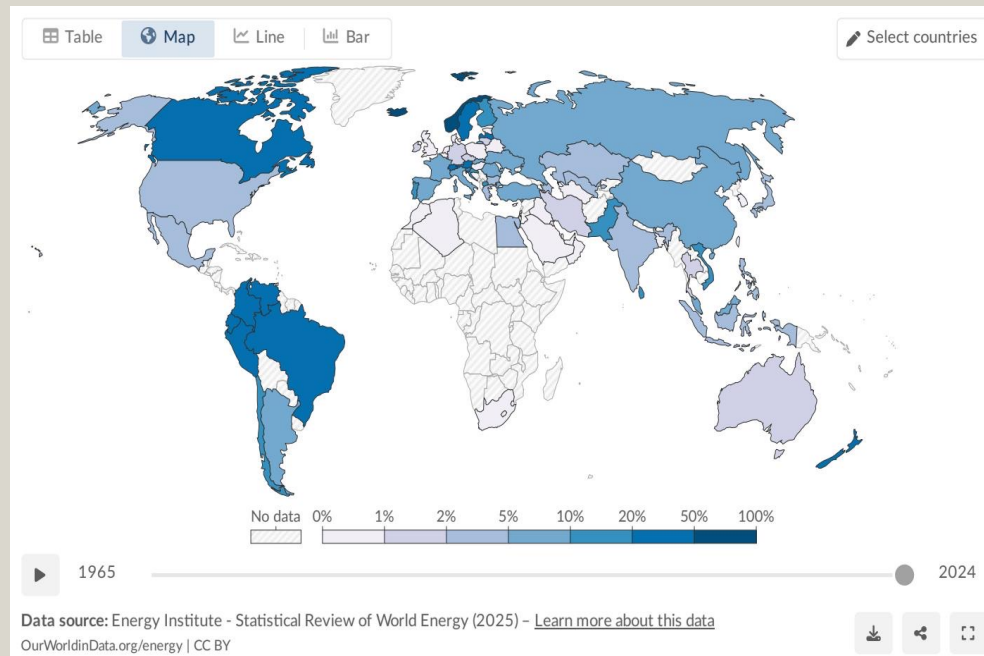
Renewables Now Supply ~14% of Global Primary Energy

Approximately one-seventh of the world's primary energy now comes from renewable technologies (2024 data, substitution method).

~14%
of global primary energy from renewables (2024)

- Primary energy encompasses electricity, transport, and heating combined
- Transport and heating are harder to decarbonize — more reliant on oil and gas
- Renewable share in total energy is lower than in electricity alone

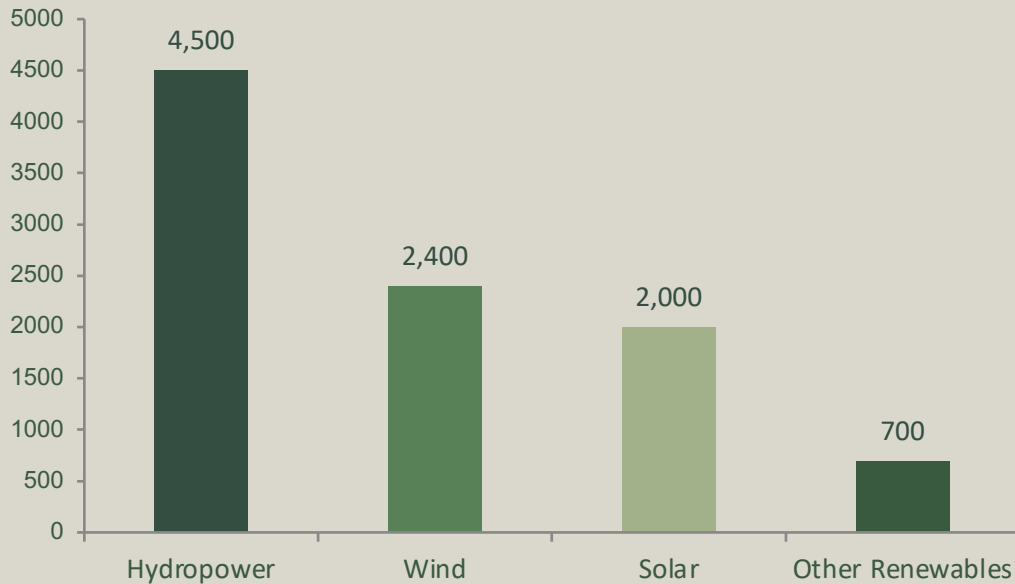
Regional leaders: Norway, Brazil, New Zealand derive large shares from renewables; Middle East and parts of Asia remain below 5%.



Share of primary energy from renewables by country, 2024

Source: Our World in Data; Energy Institute (2025)

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar



Technology Breakdown (2024 Generation)

Technology	TWh	Share
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other	~700	~7%

Wind and solar combined now nearly match hydropower's output and are growing much faster.

Wind and solar have accelerated dramatically since ~2010, driven by falling costs and policy support.

Source: Ember (2026); Energy Institute (2025)

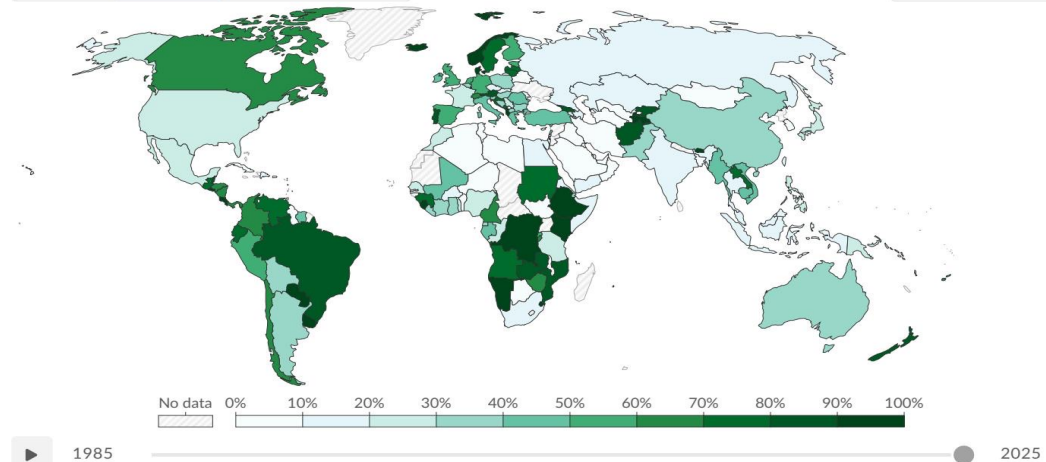
Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data

Table Map Line Bar Select countries



Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) - [Learn more about this data](#)
OurWorldinData.org/energy | CC BY

Share of electricity from renewables by country, 2025

Renewable Electricity Share by Country / Region

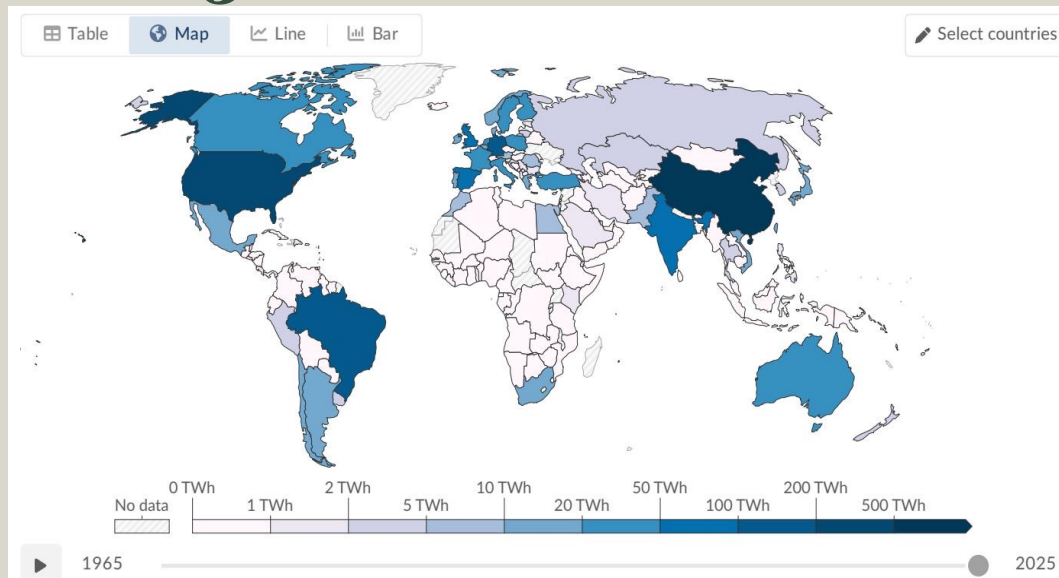
Country	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%
World Avg.	~30%

Several African and Middle Eastern countries have renewable electricity shares below 10%, reflecting fossil fuel dependence and limited investment.

Source: Ember (2026); Energy Institute (2025)

Hydropower Remains the Largest Renewable Source

~~at 50%~~ of Generation



Hydropower generation by country, 2025

~50%

of all renewable electricity
generation comes from hydropower

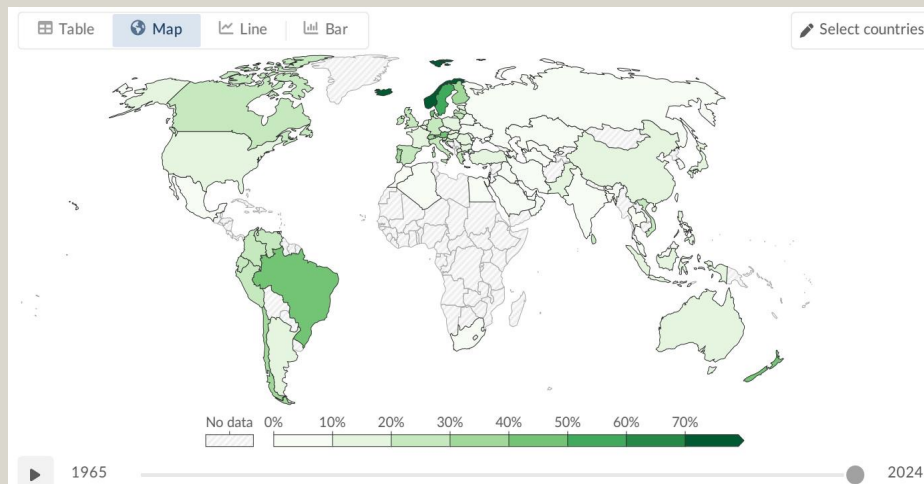
Top Producers

- China
- Brazil
- Canada

Hydropower has been generating electricity at scale for over a century. Its growth potential is limited compared to wind and solar.

Source: Ember (2026); Energy Institute (2025)

Wind and Solar Are the Fastest-Growing Energy Technologies in History



Wind power generation by country, 2025

Wind
~2,400

TWh

~25% of renewable electricity

Major in Europe, China, US

Solar
~2,000

TWh

~21% of renewable electricity

Fastest-growing source; dramatic growth since 2010

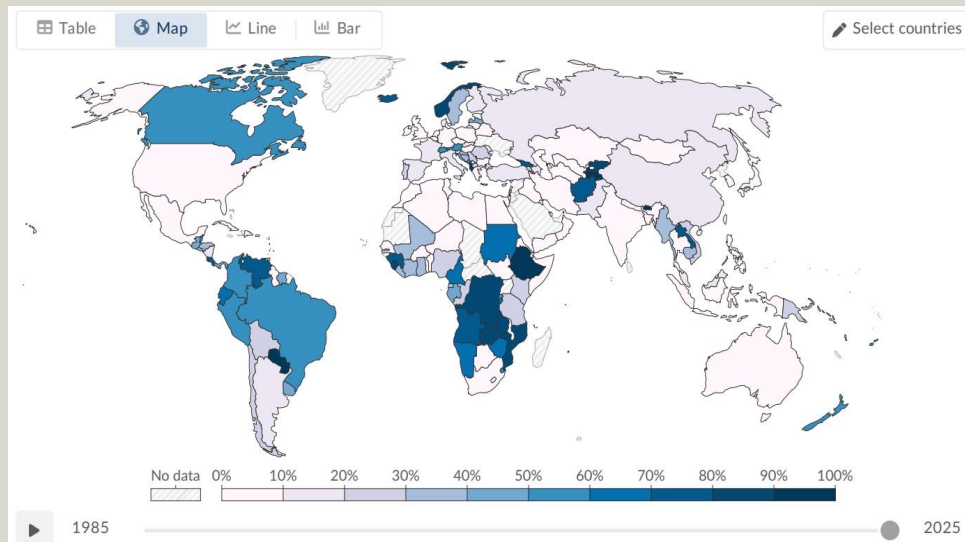
Wind + Solar Combined

~4,400 TWh | ~46% of renewable electricity — nearly matching hydropower

Driven by falling costs and policy support, wind and solar have accelerated dramatically since 2010.

Source: Ember (2026); Energy Institute (2025)

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



Share of electricity from hydropower by country, 2025

Leading Regions

- Scandinavia & Brazil: 65–98% renewable electricity
- Abundant hydropower resources
- Denmark & Germany: heavy wind & solar investment

Lagging Regions

- Sub-Saharan Africa: minimal renewable infrastructure
- Middle East: almost entirely fossil-fuel dependent
- Many countries below 10% renewable electricity

The transition to renewables is highly uneven — geography, infrastructure, and investment all play critical roles in determining progress.

Source: Ember (2026); Energy Institute (2025)

Key Takeaways

Renewables Are Growing Fast but the Transition Must Accelerate

- 1 ~14% of primary energy**
and ~30% of global electricity now come from renewables — a significant milestone but far from sufficient for climate targets
- 2 Hydropower remains dominant**
generating roughly half of all renewable electricity, but its growth potential is limited compared to wind and solar
- 3 Wind and solar are the fastest-growing**
energy technologies in history; combined, they now produce nearly as much electricity as hydropower
- 4 Stark regional disparities persist**
Scandinavia and Latin America exceed 60–90% renewable electricity, while much of the Middle East and Sub-Saharan Africa remain below 10%
- 5 Transport and heating lag behind**
The pace of the renewable transition must accelerate dramatically to meet climate goals and close regional gaps.